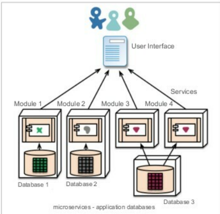

Figure 1: Microservices - application databases

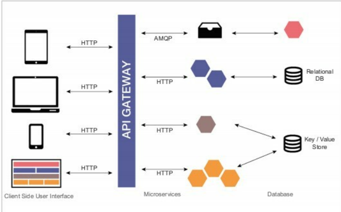

Figure 2: Microservices architecture

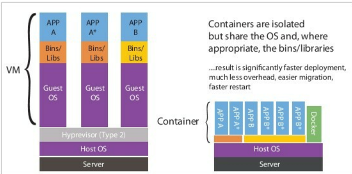

Figure 3: Virtual machines vs containers

What's so good about microservices?

With the advances in software architecture, microservices have emerged as a different platform compared to other software architecture. Microservices are easily scalable and are not limited to a language; so you are free to choose any language for the services. The services are loosely coupled, which in turn results in ease of maintenance and flexibility, as well as reduced time in debugging and deployment.

Microservices with Docker and Kubernetes

Docker is a software technology that provides containers, which are a computer virtualisation method in which the kernel of an operating system allows the existence of multiple isolated user-space instances, instead of just one. Everything required to make a piece of software run is packaged into isolated containers. With microservices, containers play the same role of providing virtual environments to different processes that are running, being deployed and undergoing testing, independently.

Docker is a bit like a virtual machine, but rather than creating a whole virtual operating system, Docker allows applications to use the same kernel as the system that it's running on and only requires applications to be shipped with things not already running on the host computer. The main idea behind using Docker is to eliminate the 'works on my machine' type of problems that occur when collaborating on code with co-workers. Docker doesn't have to install and configure complex databases nor worry about switching between incompatible language toolchain versions. When an app is dockerised, that complexity is pushed into containers that are easily built, shared and run. It is a tool that is designed to benefit both developers and systems administrators.

How well does Kubernetes go with Docker?

Before starting the discussion on Kubernetes, we must first understand orchestration, which is to arrange various components so that they achieve a desired result. It also means the process of integrating two or more applications and/or services together to automate a process, or synchronise data in real-time.

The intermediate path connecting two or more services is done by orchestration, which refers to the automated arrangement, coordination and management of software containers. So what does Kubernetes do then? Kubernetes is an open source platform for automating deployments, scaling and operations of application containers across clusters of hosts, providing container-centric infrastructure. Orchestration is an idea whereas Kubernetes implements that idea. It is a tool for orchestration. It deploys containers inside a cluster. It is a helper tool that can be used to manage a cluster of containers and treat all servers as a single unit. These containers are provided by Docker.

The best example of Kubernetes is the Pokémon Go App, which runs on a virtual environment of Google Cloud, in a separate container for each user. Kubernetes uses a different set-up for each OS. So if you want a tool that will overcome Docker's limitations, you should go with Kubernetes.

To conclude, we may say that microservices is growing very fast, the reason being its features of independence and isolation which give it the power to easily run, test and be deployed. This is just a small summary of microservices, about which there is a lot more to learn. END

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